Linear polynomial-calculus and sum-of-squares degree lower bounds for tensor isomorphism

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Editorial note. Public sources contain only the corresponding official published complete author material; native TeX served as editable transcription aid. Raw native source/tar remains private. Original theorem numbering preserved with section counters. Rank and inversion consequences counted as a single result, and the alternative Random3XOR tensor lower-bound proof is not a new record.

Editorial note. References to unselected original sections are printed with their accurate original article chapter numbers; selected section headers and original labels remain in place. No absent section is presented as an included proof.

Allowed standard prerequisites (editorial).

- Known graph-isomorphism and functional-pigeonhole polynomial-calculus lower bounds; effective Nullstellensatz and tensor-rank inheritance. Exact original cited statements retained where used; standard earlier literature not recursively reproved

Abstract

The Tensor Isomorphism problem (TI) has recently emerged as having connections to multiple areas of research within complexity and beyond, but the current best upper bound is essentially the brute force algorithm. Being an algebraic problem, TI (or rather, proving that two tensors are *non*-isomorphic) lends itself very naturally to algebraic and semi-algebraic proof systems, such as the Polynomial Calculus (PC) and Sum of Squares (SoS). For its combinatorial cousin Graph Isomorphism, essentially optimal lower bounds are known for approaches based on PC and SoS (Berkholz & Grohe, SODA '17). Our main results are an $\Omega(n)$ lower bound on PC degree or SoS degree for Tensor Isomorphism, and a nontrivial upper bound for testing isomorphism of tensors of bounded rank.

We also show that PC cannot perform basic linear algebra in sublinear degree, such as comparing the rank of two matrices, or deriving $BA = I$ from $AB = I$. As linear algebra is a key tool for understanding tensors, we introduce a strictly stronger proof system, PC+Inv, which allows as derivation rules all substitution instances of the implication $AB = I \to BA = I$. We conjecture that even PC+Inv cannot solve TI in polynomial time either, but leave open getting lower bounds on PC+Inv for any system of equations, let alone those for TI. We also highlight many other open questions about proof complexity approaches to TI.

1 Introduction

Tensors have rapidly emerged as a fundamental data structure and key mathematical object of the 21st century. They play key roles in many different areas of science, engineering, and mathematics, from quantum mechanics and general relativity to neural networks [NPOV15] and mechanical engineering. They arise in theoretical computer science in many ways, including from (post-quantum) cryptography [Pat96, JQSY19], derandomization, Matrix Multiplication, Graph Isomorphism [GQ21b], and several different parts of Geometric Complexity Theory.

The fundamental notion of equivalence between tensors is that of isomorphism: two tensors are isomorphic if one can be transformed into the other by an invertible linear change of basis in each of the corresponding vector spaces. For example, two 2-tensors (=matrices) M, M' are equivalent under this notion if there are invertible matrices X, Y such that $XMY = M'$; similarly, two 3-tensors, represented by 3-way arrays T_{ijk}, T'_{ijk} are isomorphic if there are three invertible matrices X, Y, Z such that

$$\sum_{ijk} X_{ii'} Y_{jj'} Z_{kk'} T_{ijk} = T'_{i'j'k'} \tag{1}$$

for all i', j', k'. The problem of (3-)Tensor Isomorphism (TI) is: given two such 3-way arrays, to decide if they are isomorphic.

Over finite fields, two different versions of TI sandwich the complexity of its more famous cousin, Graph Isomorphism. Namely, as presented above, GI reduces to TI. In the other direction, over a finite field $\mathbb{F} = \mathbb{F}_{p^a}$, one can take an $n \times n \times n$ tensor and list it out "verbosely", as a set of p^{an} many $n \times n$ matrices over $\mathbb{F}$; the isomorphism problem for such verbosely given tensors is equivalent to Group Isomorphism for a certain class of p-groups, widely believed to be the hardest cases of Group Isomorphism in general. As such, this verbose version of TI reduces to GI. Furthermore, with Babai's quasi-polynomial-time algorithm [Bab16], the running times are quite close: $N^{O(\log N)}$ for VerboseTI and $N^{O(\log^2 N)}$ for GI (the exponent of the exponent was worked out by Helfgott [Hel19]). Thus TI stands as a key obstacle to putting GI into P.

In this paper, we initiate the study of (algebraic) proof complexity approaches to proving that two tensors are non-isomorphic. Lower bounds on the Polynomial Calculus proof system imply lower bounds on Gröbner basis techniques, and the latter are some of the leading methods for solving TI-complete problems in cryptanalysis, e.g., [TDJ+22, FP06]. In the context of GI, proof complexity plays an important role, through its connection with the Weisfeiler–Leman (WL) algorithm. Although this algorithm does not, on its own, solve GI in polynomial time [CFI92], it is a key subroutine in many of the best algorithms for GI, both in theory [Bab16] and in practice (see [McK81, MP14]). And the picture that has emerged is that some proof systems for GI are known to be equivalent in power to WL [AM13, BG15], and some lower bounds on proof systems are closely related to lower bounds for WL [SSC14, OWWZ14, BG15]. Versions of WL for groups, and in particular finite p-groups—and hence, by the connection above, tensors over finite fields—have only recently begun to be explored [BGL+19, BS20, BS22, CL22].

1.1 Main results

We focus on the Polynomial Calculus (PC, or Gröbner) proof system [CEI96], though our results will also hold for semi-algebraic proof systems such as Sum-of-Squares [Las01] as well. PC is used to show that a system of polynomial equations over a field $\mathbb{F}$ is unsatisfiable over the algebraic closure $\overline{\mathbb{F}}$, by deriving from the system of equations, in a line-by-line fashion, the contradiction $1 = 0$. The degree of a PC proof is the maximum degree of any line appearing in the proof, and it is a fundamental result that PC proofs of constant degree can be found in polynomial time [CEI96]. Much as WL informally "captures all combinatorial approaches" to GI, PC informally "captures all approaches based on Gröbner bases" to showing that a system of polynomial equations is unsatisfiable.

The systems of equations we study are, for two non-isomorphic tensors T, T', the equations (1) along with new matrices X', Y', Z', and equations saying that these are the inverses of X, Y, Z, resp., viz.: $XX' = X'X = \mathrm{Id}$, and similarly for the others. The reason for introducing these new matrices, despite their not appearing in (1), is that these invertibility equations are only degree 2. In contrast, if we instead used the determinant to indicate that X was invertible, then our starting equations would have degree $n+1$, rather than constant degree ≤ 3. Since the main complexity measure we study on PC is degree, having starting equations of degree n would make it difficult to make meaningful lower bound statements.

Our first main result is (two proofs of) a lower bound on such techniques.

Theorem 1.1. *Over any field, there are instances of $n \times n \times n$ Tensor Isomorphism that require PC degree $\Omega(n)$ to refute. Over $\mathbb{R}$, they also require Sum-of-Squares degree $\Omega(n)$ to refute.*

The preceding goes by reduction from known lower bounds on PC for Graph Isomorphism [BG15, BG17], but has the disadvantage (from the tensor point of view) that the resulting tensors are quite sparse: in one direction, one of the slices is supported on an $\Omega(n) \times n$ matrix and all the others slices have support size 1. In a second proof (Section 6 (in the original article)), we get a polynomially worse lower bound $\Omega(\sqrt[4]{n})$, but with a reduction from Random 3XOR that is more direct. Indeed, we show that 3XOR itself can be viewed as a particular instance of a tensor problem *without* gadgets; gadgets are only then needed to reduce from that tensor problem to Tensor Isomorphism itself. In contrast, the lower bounds on PC for GI (*ibid.*) already use the Cai–Fürer–Immerman gadgets [CFI92] to reduce from XOR-SAT, and then even further gadgets are needed to reduce from GI to TI.

Our technical contributions in the above theorem are thus three-fold:

1. We show that the known reductions from GI to TI can be carried out in low-degree PC;
2. We realize 3XOR very naturally as a tensor problem; and
3. We give new reductions from 3XOR, through a series of tensor-related problems, to TI, that work as many-one reductions of the decision problems that can be carried out in low-degree PC.

Complementing our lower bound, we also show that tensors of low rank are comparatively easy to test for (non)-isomorphism. Here, one of our upper bounds is in the weaker Nullstellensatz proof system (giving a stronger upper bound than only a PC upper bound). In the Nullstellensatz proof system, a proof that a system of equations $f_1 = \cdots = f_m = 0$ is unsatisfiable consists of polynomials g_i such that $\sum g_i f_i = 1$, and the Nullstellensatz degree is the maximum degree of any $g_i f_i$. The PC degree is always at most the Nullstellensatz degree, and the gap between the two can be nearly maximal for Boolean equations: $O(1)$ versus $\Omega(n/\log n)$ [BCIP02]. (For Boolean equations, there is always an $O(n)$ upper bound, though this does not apply to TI, see Remark 1.3 below).

Theorem 1.2. *Over any field, the Nullstellensatz degree of refuting isomorphism of two $n \times n \times n$ tensors of tensor rank $\leq r$ is at most $2^{O(r^2)}$. If working over a finite field $\mathbb{F}_q$ and including the equations $x^q - x$, the PC degree is at most $O(qr^2)$.*

In particular, isomorphism of constant-rank tensors can be decided in polynomial time.

Remark 1.3. In many settings in proof complexity, Boolean axioms such as $x_i^2 = x_i$ or $x_i^2 = 1$ are included among the system of equations, and all such unsatisfiable systems of equations can be refuted in degree $O(n)$ ($n =$# variables). If this were the case here, the above would only be interesting for very small values of r. In contrast, the equations for TI do not include any such Boolean axioms, and as such the naive degree upper bound is exponential in the number of variables. For $n \times n \times n$ tensors, this gives an upper bound of $2^{O(n^2)}$ [Som99], and thus, Theorem 1.2 gives nontrivial upper bounds all the way up to $r \leq n$. (We note that $n \times n \times n$ tensors can have rank up to $\Theta(n^2)$ [Lic85].) The proof of Theorem 1.2 shows that for rank-r tensors, TI can essentially be reduced to a system of equations in only $O(r^2)$ variables.

Remark 1.4. For fixed r, testing if an $n \times n \times n$ tensor has rank $\leq r$ can be done in polynomial time, as follows. This will show that the algorithm of Theorem 1.2 genuinely solves the decision problem, and not just a promise problem. Given an $n \times n \times n$ tensor T, consider its three $n \times n^2$ flattenings. Use Gaussian elimination to put each such flattening, separately, into reduced row echelon form. If any of these flattenings has rank $> r$, reject. Otherwise, we get from this a list of $3r$ vectors $u_1, \ldots, u_r, v_1, \ldots, v_r, w_1, \ldots, w_r$, such that T lives in the $r \times r \times r$-dimensional space $\mathrm{Span}\{u_1, \ldots, u_r\} \otimes \mathrm{Span}\{v_1, \ldots, v_r\} \otimes \mathrm{Span}\{w_1, \ldots, w_r\}$. Now in this space we can write down the Brent equations [Bre70] for T to have rank $\leq r$, which will be r^3 cubic equations in $3r^2$ variables (Brent's equations [Bre70, (5.06)] were specifically for the matrix multiplication tensor, but analogous equations are easily constructed for arbitrary tensors using the same idea). Since r is constant, these equations may be solved in polynomial time (here we assume that we are either working over a finite field, a finite-degree extension of the rationals—see, for example, Grigoriev [Gri13]—or in the BSS model over an arbitrary field).

Lastly, one may wonder why we focus on 3-Tensor Isomorphism, and not some of its many related variants. Indeed, just as there are other equivalence notions for matrices—such as conjugacy XMX^{-1} and congruence XMX^T—there are many different kinds of multilinear objects that can be represented by

multi-way arrays, including tensors, homogeneous polynomials (commutative or noncommutative), alternating matrix spaces, multilinear maps, and so on, each with their own corresponding notion of isomorphism. While these problems are indeed distinct, they are all equivalent under polynomial-time isomorphisms [FGS19, GQ21b]; such problems are called TI-complete. Even isomorphism of k-way tensors (for any fixed $k \geq 3$) is equivalent to isomorphism of 3-tensors [GQ21b]. This partially justifies our focus on 3-Tensor Isomorphism. In the course of proving our reductions for the results stated above, we use many of the gadgets from [FGS19, GQ21b], and show that such uses also often yield proof complexity reductions as well. Because of the variety of gadgets used in our reductions, we believe that many, if not all, of the gadgets from those results would also yield proof complexity reductions, so the proof complexity of all the known TI-complete problems should be polynomially related.

1.2 Comparison with linear algebra, a new proof system, and a conjecture

As linear algebra is part of the core toolkit for understanding tensors, it is natural to wonder how linear algebra can help in algebraic proof complexity approaches to TI. We believe that even if it had the "full power" of linear algebra at its disposal "for free," PC could still not solve TI efficiently. We begin to make this precise in this section.

Some basic derivations in linear algebra are to relate the ranks of two matrices and to derive $BA = I$ from $AB = I$ (the Inversion Principle, one of the so-called "hard matrix identities" [SC04], only recently shown to have short NC^2-Frege proofs [HT15]). Soltys [Sol01] and Soltys & Cook [SC04] discuss the relationship between these and other standard implications in linear algebra. We show that PC is not strong enough to prove these in low-degree:

Theorem 1.5. *The unsatisfiable system of equations $XY = \mathrm{Id}_n$ where X is $n \times r$ and Y is $r \times n$ with $1 \leq r < n$, requires degree $\geq r/2 + 1$ to refute in PC, over any field.*

We refer to this system of equations as the Rank Principle, as refuting them amounts to showing that $\mathrm{rk}\,\mathrm{Id}_n > r$.

Theorem 1.6. *Any PC derivation of $BA = I$ from $AB = I$, where A, B are $n \times n$ matrices with $\{0, 1\}$ entries, requires degree $\geq n/2 + 1$, over any field.*

We also observe that the Rank Principle can be derived in low degree from the Inversion Principle.

Although it remains open whether the Inversion Principle is "complete" for linear-algebraic reasoning (see [Sol01, SC04]), we introduce the proof system PC+Inv in an attempt to capture some linear-algebraic reasoning that seems potentially useful for TI. PC+Inv has all the same derivation rules as PC, but in addition, for any square matrices A, B (whose entries may themselves be polynomials—that is, we allow substitution instances), we have the rule

$$\frac{AB = I}{BA = I}.$$

where the antecedent represents the set of n^2 equations corresponding to $AB = I$, and similarly the consequent denotes the set of n^2 equations $BA = I$ (see

2.3 for more details). Degree is still measured in the usual way, but this rule lets us "cut out" the high-degree proof that would usually be required to derive $BA = I$ from $AB = I$. We now formalize our intuition that linear algebra should not suffice to solve TI efficiently in the following:

Conjecture 1.7. Tensor Isomorphism for $n \times n \times n$ tensors requires degree $\Omega(n)$ in PC+Inv, over any field.

Despite the conjecture, we do not yet know how to prove lower bounds on PC+Inv for any unsatisfiable system of equations, let alone those coming from TI. Mod p counting principles (for p different from the characteristic of the field) strike us as potentially interesting instances to examine for PC+Inv lower bounds, before tackling a harder problem like TI. In the final section, we highlight many other open questions around the proof complexity of TI.

1.3 Organization

In Section 2 we cover preliminaries. In Section 3 (in the original article) we prove the lower bounds on linear algebraic principles just discussed. In Section 4 (in the original article) we prove the upper bound for isomorphism of bounded rank tensors (Theorem 1.2). In Section 5 we prove Theorem 1.1 by reduction from GI. In Section 6 (in the original article) we prove the polynomially related lower bound by direct reduction from Random 3XOR.

2 Preliminaries

2.1 Proof systems

All our rings are commutative and unital. *Polynomial calculus* (PC) is a proof system to prove that a given system of (multivariate) polynomial equations $\mathcal{P}$ over a field $\mathbb{F}$ of the form $p = 0$, has no solution over the algebraic closure $\overline{\mathbb{F}}$ (i.e. the system is unsolvable). We usually shorten the polynomial equation $p = 0$ to just p. The derivation rules of the system are the following one:

$$\frac{p}{xp} \text{ (multiplication)}, \quad \frac{p \quad q}{ap + bq} \text{ (linear combination)}$$

where x is any formal variable, $a, b \in \mathbb{F}$ and p, q are polynomials over $\mathbb{F}$.

When refuting Boolean systems of equations it is common to include the Boolean axioms $x_i^2 - x_i$. Because we do *not* always include these (esp. for TI) we are explicit about our use of these, but do not assume they are built into the proof system—that is, if we are assuming them as axioms, we say so.

A *PC derivation (or proof)* of a polynomial q from a set of polynomials $\mathcal{P}$ is a sequence of polynomial equations $p_1, \ldots, p_m$ ending with the polynomial q (so p_m is q) and where each p_i, $i \in [m]$, is either an *axiom* p for $p \in \mathcal{P}$, or is obtained from previous equations in the refutation by multiplication or linear combination. We denote this by writing $\mathcal{P} \vdash q$. Observe that if p is derivable in PC and q is a polynomial then, by repeated applications of multiplication and linear combination rules, we can derive pq. We often use this generalization of the multiplication in our proofs without mention.

A *PC refutation* is just a PC proof of the polynomial 1. The *degree* of a PC derivation is the maximal degree of a polynomial used in the proof. The *size* of

a polynomial p is the number of terms in p. The *size* a PC derivation $p_1, \ldots, p_m$ is the sum of the sizes of the polynomials $p_1, \ldots, p_m$.

For our upper bound in Theorem 1.2, we also consider another algebraic proof system, known as *Nullstellensatz* (NS), to certify unsolvability of sets of polynomial equations. Nullstellensatz is defined in a static form as follows: a refutation of a list $\mathcal{P} = (p_1, \ldots, p_m)$ of polynomial equations over variables $x_1 \ldots, x_n$ is given by the list of polynomials $\mathcal{Q} = (q_1, \ldots q_m)$ such that

$$\sum_{i \in [m]} p_i q_i = 1$$

The *degree* of a NS refutation is the maximal degree of a polynomial in $\mathcal{P} \cup \mathcal{Q}$. The size of NS proof is the sum of the number of monomials appearing in the polynomials $q_1, \ldots, q_m$.

Sum-of-Squares (SOS) is a static proof system for certifying the unsolvability of systems of polynomial equations and polynomial inequalities, where polynomials are usually over the ring $\mathbb{R}[x_1 \ldots, x_n]$.

A polynomial p is a *sum-of-squares* polynomial if it is in the form $p = \sum_i r_i^2$ and the r_i's are polynomials as well. Given a system made by a set of polynomial equations $\mathcal{P} = \{p_1 = 0, \ldots p_m = 0\}$ and a set $\mathcal{Q} = \{q_1 \geq 0, \ldots q_k \geq 0\}$ of polynomial inequalities, a *sum-of-squares* proof of the polynomial inequality $p \geq 0$ from $\mathcal{P} \cup \mathcal{Q}$ is given by the formal identity

$$p = s_0 + \sum_{i \in [k]} s_i q_i + \sum_{j \in [m]} t_j p_j$$

where $s_0, s_1, \ldots, s_k$ are sum-of-squares polynomials, while $t_1, \ldots, t_m$ are arbitrary polynomials. When the system $\mathcal{P} \cup \mathcal{Q}$ is unsatisfiable, a *refutation* of $\mathcal{P} \cup \mathcal{Q}$ is a proof of the inequality $-1 \geq 0$, that is for p the constant polynomial -1. The *degree* of the proof is the $\max\{\deg(p), \deg(s_0), \deg(s_i) + \deg(q_i), \deg(t_j) + \deg(p_j) | i \in [k], j \in [m]\}$.

Definition 2.1 (PC reduction between systems of polynomials, cf. [BGIP01, Sec. 3])**.** Let $P(x_1, \ldots, x_n)$ and $Q(y_1, \ldots, y_m)$ be two sets of polynomials over a field $\mathbb{F}$. P is (d_1, d_2)-*reducible* to Q if:

1. For each $i \in [m]$ there is a polynomial $r_i(\mathbf{x})$ of degree at most d_1 (which we think of as defining y_i in terms of the $\mathbf{x}$ variables);
2. There exists a degree d_2 PC derivation of $Q(r_1(\mathbf{x}), \ldots, r_m(\mathbf{x}))$ from polynomials $P(\mathbf{x})$.

Lemma 2.2 ([BGIP01, Lem. 1])**.** *If $P(\mathbf{x})$ is (d_1, d_2)-reducible to $Q(\mathbf{y})$ and there is a degree d PC refutation of $Q(\mathbf{y})$, then there is a degree $\max(d_2, d_1 d)$ refutation of $P(\mathbf{x})$.*

In their paper, they typically only applied this to systems of equations which were known to be unsatisfiable (such as PHP and Tseitin tautologies), whereas in our paper we have several situations we want to combine the above notion together with the usual notion of many-one reduction. We encapsulate this in

the following definition. We say a decision problem Π is a *polynomial solvability problem* over a field $\mathbb{F}$ if all valid instances of the problem are systems of polynomial equations over $\mathbb{F}$, and the problem is to decide whether such a system of equations has solutions over the algebraic closure $\overline{\mathbb{F}}$. Thus, the difference between multiple polynomial solvability problems is just *which* systems of equations are valid inputs.

Definition 2.3 (PC many-one reduction)**.** Let Π_1, Π_2 be two polynomial solvability problems over a field $\mathbb{F}$. We say that Π_1 (d_1, d_2)-*many-one reduces* to Π_2 if there is a polynomial-time many-one reduction ρ from Π_1 to Π_2, such that for all unsatisfiable instances $\mathcal{F}$ of Π_1, $\mathcal{F}$ (d_1, d_2)-reduces to $\rho(\mathcal{F})$. When this occurs with $d_1, d_2 = O(1)$, we write

$$\Pi_1 \leq_m^{PC} \Pi_2.$$

2.2 Linear algebra and tensors

Given three vector spaces U, V, W over a field $\mathbb{F}$, a 3-tensor is an element of the vector space $U \otimes V \otimes W$, whose dimension is $(\dim U)(\dim V)(\dim W)$. If e_i is the i-th standard basis vector, then a basis for $U \otimes V \otimes W$ is given by the vectors $\{e_i \otimes e_j \otimes e_k\}$. One may also interpret the symbol $\otimes$ more concretely as the Kronecker product, in which $e_i \otimes e_j \otimes e_k$ represents a 3-way array whose only nonzero entry is in the (i, j, k) position. The vector space of such 3-way arrays (with coordinate-wise addition) is isomorphic to $U \otimes V \otimes W$.

The rank of a tensor $T \in U \otimes V \otimes W$ is the minimum r such that $T = \sum_{i=1}^{r} u_i \otimes v_i \otimes w_i$ for some vectors u_i, v_i, w_i.

Two $n \times m \times p$ 3-tensors $T, T' \in U \otimes V \otimes W$ are isomorphic if there exist matrices $X \in \mathrm{GL}(U), Y \in \mathrm{GL}(V), Z \in \mathrm{GL}(W)$ such that $(X, Y, Z) \cdot T = T'$, where the latter is shorthand for (1). If we treat T, T' as given non-isomorphic tensors, then we may treat (1) as a system of equations in the $n^2 + m^2 + p^2$ variables $X_{ii'}, Y_{jj'}, Z_{kk'}$. To enforce that these variable matrices are invertible, we furthermore introduce three additional sets of variables X', Y', Z' meant to be the inverse matrices, and include also the equations

$$XX' = X'X = I_n \qquad YY' = Y'Y = I_m \qquad ZZ' = Z'Z = I_p,$$

where I_n denotes the $n \times n$ identity matrix, which is Id_U in any basis. (We could have instead introduced new variables such as δ and the equation $\det(X)\delta = 1$, however, the latter equation is degree n, whereas the above equations all have degree $O(1)$, which is more desirable from the point of view of algebraic proof complexity.)

2.3 Polynomial encodings and the inversion principle

Some principles of linear algebra can be formulated as tautologies in propositional logic and therefore also as a set of polynomial equations. In this paper we preliminarily consider two such principles.

Rank Principle. As a first example we consider a set of unsatisfiable polynomials encoding the principle that the product of a $n \times r$ matrix X by a $r \times n$ matrix Y cannot be the identity matrix whenever $r < n$. We consider variables

$x_{i,k}, y_{j,k}$ for $i, j \in [n]$ and $k \in [r]$, where $r < n$ to encode X and Y. Then the polynomial encoding is:

$$\mathbb{I}(r,n) := \sum_{k\in[r]} x_{i,k}y_{j,k} - \delta_{i,j} \qquad i,j \in [n]$$

where $\delta_{i,j} = 1$ if $i = j$ and 0 otherwise. This set of polynomials is clearly unsatisfiable as long as $r < n$.

Inversion Principle. The second principle encodes the invertibility of a square $n \times n$ matrix A, expressing the tautology that $AB = I \to BA = I$ where A, B are $n \times n$ matrices and I is the identity matrix. Stephen A. Cook suggested this principle as a tautology that may be hard to prove in several proof systems.

Let $a_{i,j}, b_{i,j}$ be formal variables encoding respectively the (i,j)-th entries of A and B. We represent the fact that $AB = I$ as the set of degree 2 polynomials

$$\sum_{k\in[n]} a_{i,k}b_{k,j} - \delta_{i,j} \qquad i,j \in [n],$$

where $\delta_{i,j} = 1$ if $i = j$ and 0 otherwise. We denote this set of polynomials by $AB = I$. In Section 3 (in the original article), we study the degree complexity of $AB = I \vdash BA = I$, that is of PC derivations of the polynomials $BA = I$ from the polynomials $AB = I$.

In view of the results we obtain in Section 3 (in the original article), in Section 1.2 we considered a *polynomial rule schema* of the form

$$\frac{AB = I}{BA = I}$$

which we call the *Inversion Rule* (INV) meant to be added to PC as an extra rule. We make this slightly more precise here.

A polynomial instantiation τ of the polynomials $AB = I$ is a substitution of polynomials $p_{i,j}, q_{i,j}$ to variables $a_{i,j}$ and $b_{i,j}$. In PC+INV a polynomial p is derivable from a set of polynomials $\mathcal{P}$ if

1. p is an axiom, or $p \in \mathcal{P}$;
2. p is obtained by multiplication or linear combination from previous polynomials in the proof;
3. p is a polynomial among a polynomial instantiation τ of $BA = I$, given that among the polynomials previously derived in the proof there are all the polynomials forming the instantiation τ of $AB = I$.

Pigeonhole Principle. An important role in proving the results in Section 3 (in the original article) is played by the well-known *Pigeonhole principle* stating that any function f from $[n]$ to $[r]$ with $r < n$ has a collision, that is there are $i \neq i' \in [n]$ and a $j \in [r]$ such that $f(i) = f(i') = j$. PHP_r^n is the set of polynomials:

$$\sum_{k\in[r]} p_{i,k}{-}1, \text{ for } i \in [n], \qquad p_{i,k}p_{j,k}, \text{ for } i \neq j \in [n], k \in [r] \qquad p_{ij}^2{-}p_{ij}, \text{ for } i \in [n], j \in [r]$$

Razborov [Raz98] additionally included the "functional equations" (encoding that each pigeon cannot be matched to more than one hole):

$$p_{i,k}p_{i,k'}, \text{ for } i \in [n], k \neq k' \in [r].$$

5 Lower bound on PC degree for Tensor Isomorphism from Graph Isomorphism

Definition 5.1. Given two graphs G, H with adjacency matrices A, B (resp.), the equations for Graph Isomorphism (the same as those used by Berkholz & Grohe [BG15, BG17]) are as follows. Let Z be an $n \times n$ matrix of variables z_{ij} (where the intended interpretation is that $z_{ij} = 1$ iff an isomorphism maps vertex $i \in V(G)$ to vertex $j \in V(H)$). We say that a partial map, which sends $(i, i') \mapsto (j, j')$ is a local isomorphism if (1) $i = i'$ iff $j = j'$ (it's a well-defined map) and (2) $(i, i') \in E(G) \Leftrightarrow (j, j') \in E(H)$. (One may also do Colored Graph Isomorphism and require that the colors match, $c(i) = c(j), c(i') = c(j')$.) Then the equations are:

$z_{ij}^2 - z_{ij}$	$\forall i, j$	All variables $\{0, 1\}$-valued
$1 - \sum_i z_{ij}$	$\forall j$	each $j \in V(H)$ is mapped to from exactly one vertex
$1 - \sum_j z_{ij}$	$\forall i$	each $i \in V(G)$ maps to exactly one vertex
$z_{ij}z_{i'j'}$		Whenever $(i, i') \mapsto (j, j')$ is not a local isomorphism.

In this section, we prove a lower bound on PC (and SoS) for TI, by reducing from GI and using the known lower bounds on GI [BG15, BG17]. Specifically, we show

Theorem 5.2. *Over any field, there are instances of Tensor Isomorphism of size $O(n) \times O(n) \times O(n)$ that require PC degree $\Omega(n)$ to refute. The same holds over the reals for SoS degree.*

Proof. Berkholz and Grohe [BG15, BG17] show the same statement for n-vertex graphs of bounded vertex degrees, with the same PC/SoS degree bound. In Proposition 5.4 we show that GI reduces to Monomial Code Equivalence by a (2,4)-many-one reduction that turns n-vertex, m-edge graphs into $m \times (3m+n)$ matrices. in Proposition 5.5 we show that Monomial Code Equivalence reduces to TI by a (2,4)-many-one reduction that turns $k \times N$ matrices into $(k + 2N) \times N \times (1 + 2N)$ tensors. By Lemma 2.2, this completes the proof. □

To reduce from GI to TI we use the following intermediate problem. A matrix is *monomial* if it has exactly one nonzero entry in each row and column; equivalently, a monomial matrix is the product of a permutation matrix and an invertible diagonal matrix.

Definition 5.3. Monomial Code Equivalence is the problem: given two $k \times n$ matrices C, C', do there exist matrices X, Y such that $XCY^T = C'$ where X is invertible and Y is invertible and monomial? Given two such matrices C, C', the equations for Monomial Code Equivalence are as follows. There

are $2(k^2+n^2)$ variables arranged into matrices X, X' (of size $k \times k$) and Y, Y' (of size $n \times n$). The equations are

$$XCY^T = C' \qquad XX' = X'X = \mathrm{Id} \qquad YY' = Y'Y = \mathrm{Id}$$

and

$$\begin{array}{ll} y_{ij}y_{ij'} (\forall i \forall j \neq j') & y_{ij}y_{i'j} (\forall i \neq i', \forall j) \\ y'_{ij}y'_{ij'} (\forall i \forall j \neq j') & y'_{ij}y'_{i'j} (\forall i \neq i', \forall j) \end{array}$$

(Note: there are no equations forcing the variables to take on values in $\{0,1\}$.)

Proposition 5.4. *The reduction of Petrank & Roth [PR97] from* Graph Isomorphism *to* Linear Code Equivalence *over* $\mathbb{F}_2$ *in fact gives a (2,4)-many-one reduction from* Graph Isomorphism *to* Monomial Code Equivalence *(sic!) over any field.*

Proof. The reduction of Petrank & Roth is as follows: given a simple undirected graph G with n vertices and m edges, let $D(G)$ be its $m \times n$ incidence matrix: $D_{e,v} = 1$ iff $v \in e$ and is 0 otherwise, and let $M(G)$ be the $m \times (3m+n)$ matrix

$$M(G) = \left[\begin{array}{c|c|c|c} I_m & I_m & I_m & D(G) \end{array} \right].$$

Many-one reduction. It was previously shown (over $\mathbb{F}_2$ in [PR97] and over arbitrary fields in [Gro12, Lem. II.4]) that this gives a many-one reduction to Permutational Code Equivalence. Here we observe that the same reduction also gives a reduction to Monomial Code Equivalence. Thus, all that remains to show is that if $M(G)$ and $M(H)$ are monomially equivalent, then G must be isomorphic to H.

In fact, what was shown in [PR97] (over arbitrary fields in [Gro12]) is that, up to permutation and scaling of the rows, $M(G)$ is the unique generator matrix of its code satisfying the following properties: (1) $M(G)$ is $m \times (3m+n)$, (2) each row has Hamming weight ≤ 5, (3) any linear combination that includes two or more rows with nonzero coefficients has Hamming weight ≥ 6.

Now, suppose (X, Y) is a monomial equivalence of the codes $M(G), M(H)$. Then the rowspans of $M(G)Y^T$ and $M(H)$ are the same. Since Y is monomial, if we consider just the supports of the rows of $M(G)Y^T$, up to re-ordering the rows, by the preceding paragraph, those supports must be the same as the supports of the rows of $M(H)$. Thus X must also be monomial. Say $X = DP$ and $Y = EQ$ where D, E are diagonal and P, Q are permutation matrices. Then $PM(G)Q^T$ has the same support as $XM(G)Y^T = M(H)$, and since P and Q are permutation matrices and $M(G)$ and $M(H)$ have all entries in $\{0,1\}$, we must have $PM(G)Q^T = M(H)$. Thus $M(G)$ and $M(H)$ are in fact equivalent by a *permutation* matrix (in place of the monomial matrix Y). Thus, by the fact that $(G, H) \mapsto (M(G), M(H))$ was a reduction to Permutational Code Equivalence, we conclude that $G \cong H$.

Low-degree PC reduction. Let X, X', Y, Y' be the variable matrices in the equations for Monomial Code Equivalence of $M(G), M(H)$, and let Z be the variable matrix in the equations for Graph Isomorphism of G, H. Let $n = |V(G)|, m = |E(G)|$; so, X, X' are of size m, Y, Y' are of size $3m+n$, and Z is of size n.

Let $Z^{(2)}$ denote the $\binom{n}{2} \times \binom{n}{2}$ matrix whose $(\{i,i'\},\{j,j'\})$ entry is $z_{ij}z_{i'j'} + z_{ij'}z_{i'j}$. The idea is that if Z is a map on the vertices, then $Z^{(2)}$ is the corresponding map on the edges; the two terms come from the fact that the edge $\{i,i'\}$ can be mapped to the edge $\{j,j'\}$ either by $(i,i') \mapsto (j,j')$ or by $(i,i') \mapsto (j',j)$. Note that, since Z is a permutation matrix, at most one of these terms is nonzero, and thus $Z^{(2)}$ is also a $\{0,1\}$-matrix (in fact, a permutation matrix). Let $Z_E^{(2)}$ denote the $|E| \times |E|$ submatrix of $Z^{(2)}$ all of whose row indices are $\{i,i'\} \in E(G)$ and all of whose column indices are $\{j,j'\} \in E(H)$. Note also that $(Z_E^{(2)})^T = (Z^T)_E^{(2)}$, so we use these notations interchangeably for convenience.

Now consider the following substitution:

$$\begin{array}{ll} X \mapsto (Z_E^{(2)})^T & Y \mapsto (Z^T)_E^{(2)} \oplus (Z^T)_E^{(2)} \oplus (Z^T)_E^{(2)} \oplus (Z^T) \\ X' \mapsto Z_E^{(2)} & Y' \mapsto Z_E^{(2)} \oplus Z_E^{(2)} \oplus Z_E^{(2)} \oplus Z \end{array}$$

After making these substitutions in the equations for Monomial Code Equivalence of $M(G), M(H)$, we get the equations

$$(Z_E^{(2)})^T Z_E^{(2)} = Z_E^{(2)}(Z_E^{(2)})^T = \mathrm{Id}_m \qquad (Z_E^{(2)})^T D(G) Z = D(H) \qquad ZZ^T = Z^T Z = \mathrm{Id}_n \tag{2}$$

along with equations saying that Z and $Z_E^{(2)}$ are monomial.

We now show how to derive these equations in low-degree PC from the GI equations.

The monomial equations for Z are part of the GI equations, so there is nothing to do for those.

The monomial equations for $Z_E^{(2)}$ are of the form $(z_{ij}z_{i'j'} + z_{ij'}z_{i'j})(z_{k\ell}z_{k'\ell'} + z_{k\ell'}z_{k'\ell})$ where either (1) $\{i,i'\} = \{k,k'\}$ and $\{j,j'\} \neq \{\ell,\ell'\}$ or (2) vice versa. We expand out to get

$$z_{ij}z_{i'j'}z_{k\ell}z_{k'\ell'} + z_{ij}z_{i'j'}z_{k\ell'}z_{k'\ell} + z_{ij'}z_{i'j}z_{k\ell}z_{k'\ell'} + z_{ij'}z_{i'j}z_{k\ell'}z_{k'\ell}$$

We show how to get this equation in case (1); case (2) follows similarly, *mutatis mutandis*. In case (1), without loss of generality suppose that $i = k$, $i' = k'$, and $j \notin \{\ell,\ell'\}$. The first two terms are divisible by the GI equations $z_{ij}z_{i\ell}$ (since $i = k$ and $j \neq \ell$), the third term is divisible by $z_{i'j}z_{i'\ell'}$ (since $i' = k'$ and $j \neq \ell'$), and the last term is divisible by $z_{i'j}z_{i'\ell}$ similarly.

Next, the equations $ZZ^T = \mathrm{Id}_n$ are, expanded out,

$$\sum_j z_{ij}z_{ij} - 1 (\forall i) \qquad \sum_j z_{ij}z_{kj} (\forall i \neq k).$$

The first is gotten by linear combination from $1 - \sum_j z_{ij}$ and the Boolean axioms $z_{ij}^2 - z_{ij}$. The second is a linear combination of the monomial axioms $z_{ij}z_{kj}$ (part of the local non-isomorphism axioms). Similarly for $Z^T Z = \mathrm{Id}$, using $1 - \sum_i z_{ij}$ instead.

Next, we expand out the equations $Z_E^{(2)}(Z^T)_E^{(2)} = \mathrm{Id}_m$, to get[1]

$$\sum_{\{j,j'\} \in E(H)} (z_{ij}z_{i'j'} + z_{ij'}z_{i'j})(z_{kj}z_{k'j'} + z_{k'j}z_{kj'}) - \delta_{\{i,i'\},\{k,k'\}} (\forall \{i,i'\},\{k,k'\} \in E(G))$$

[1]We use the notation $\sum_{\{j,j'\} \in E(H)}$ to denote a sum in the index of summation takes on the value $e \in E(H)$ for each edge of H exactly once. Because our edges are undirected, we only use such sums when the summand expression is itself invariant under swapping the roles of j, j'. If so desired, one could equivalently say $\sum_{j<j',\{j,j'\} \in E(H)}$.

Thus, for $\{i,i'\} \neq \{k,k'\}$, we need to derive

$$\sum_{\{j,j'\}\in E(H)} \left(z_{ij}z_{i'j'}z_{kj}z_{k'j'} + z_{ij'}z_{i'j}z_{kj}z_{k'j'} + z_{ij}z_{i'j'}z_{k'j}z_{kj'} + z_{ij'}z_{i'j}z_{k'j}z_{kj'}\right).$$

Without loss of generality, suppose that $i \notin \{k,k'\}$. Then the first two terms of each summand are divisible by the GI equation $z_{ij}z_{kj}$, the third term is divisible by $z_{ij}z_{k'j}$, and the last term is divisible by $z_{ij'}z_{kj'}$. On the other hand, when $\{i,i'\} = \{k,k'\}$, we need to derive

$$-1 + \sum_{\{j,j'\}\in E(H)} \left(z_{ij}^2 z_{i'j'}^2 + 2z_{ij'}z_{i'j}z_{ij}z_{i'j'} + z_{ij'}^2 z_{i'j}^2\right).$$

The middle terms of each summand are divisible by the GI equations $z_{ij'}z_{ij}$. For the first and third terms, we can use the Boolean axioms to remove the squares, and thus we are left to derive

$$-1 + \sum_{\{j,j'\}\in E(H)} (z_{ij}z_{i'j'} + z_{ij'}z_{i'j}) \tag{3}$$

We derive this from the GI equations as follows. Consider $(\sum_j z_{ij} - 1)(\sum_{j'} z_{i'j'} - 1) + (\sum_j z_{ij} - 1) + (\sum_{j'} z_{i'j'} - 1)$ and break up the resulting sum according to whether $j = j'$, $\{j,j'\} \in E(H)$ or $\{j,j'\} \notin E(H)$. Then we get

$$\sum_j z_{ij}z_{i'j} + \sum_{j,j':\{j,j'\}\in E(H)} z_{ij}z_{i'j'} + \sum_{j\neq j'\{j,j'\}\notin E(H)} z_{ij}z_{i'j'} - 1$$

Every summand in the first sum is a monomial axiom since $i \neq i'$. Every summand in the third sum is a local non-isomorphism axiom, since $\{i,i'\} \in E(G)$ but $\{j,j'\} \notin E(H)$. Note that every edge $\{j,j'\}$ of $E(H)$ is represented twice in the middle sum: once as (j,j') and once as (j',j). Thus, the above simplifies to

$$\sum_{\{j,j'\}\in E(H)} (z_{ij}z_{i'j'} + z_{ij'}z_{i'j}) - 1,$$

which is what we sought to derive. The derivation of $(Z_E^{(2)})^T Z_E^{(2)} = \mathrm{Id}$ is similar.

Finally, we show how to derive the equation $(Z_E^{(2)})^T D(G) Z = D(H)$ from the equations $ZA(G) = A(H)Z$, where $A(G)$ denotes the adjacency matrix of G, with $A(G)_{ii'} = 1$ iff $\{i,i'\} \in E(G)$. Writing out the equations in indices, we need to derive

$$\sum_{\{i,i'\}\in E(G),k\in V(G)} \left(Z_E^{(2)}\right)_{\{i,i'\},\{j,j'\}} D(G)_{\{i,i'\},k} z_{k\ell} = D(H)_{\{j,j'\},\ell} (\forall \ell \in V(H), \forall \{j,j'\} \in E(H))$$

Using the fact that $D(G)_{\{i,i'\},k} = \delta_{ik} + \delta_{i'k}$ and the definition of $Z^{(2)}$, this is the same as

$$\sum_{\{i,i'\}\in E(G),k\in V(G)} (z_{ij}z_{i'j'} + z_{ij'}z_{i'j})(\delta_{ik}+\delta_{i'k})z_{k\ell} = \delta_{j\ell}+\delta_{j'\ell} (\forall \ell \in V(H), \forall \{j,j'\} \in E(H))$$

Thus we need to derive:

$$\sum_{\{i,i'\}\in E(G)} (z_{ij}z_{i'j'} + z_{ij'}z_{i'j})(z_{i\ell} + z_{i'\ell}) = \begin{cases} 1 & \ell \in \{j,j'\} \\ 0 & \text{otherwise.} \end{cases}$$

Expanding out the summand, we find the four terms

$$z_{ij}z_{i'j'}z_{i\ell} + z_{ij}z_{i'j'}z_{i'\ell} + z_{ij'}z_{i'j}z_{i\ell} + z_{ij'}z_{i'j}z_{i'\ell}.$$

When $\ell \notin \{j, j'\}$, each of these terms is divisible by one of the monomial (local non-isomorphism) axioms, respectively: $z_{ij}z_{i\ell}$, $z_{i'j'}z_{i'\ell}$, $z_{ij'}z_{i\ell}$, and $z_{i'j}z_{i'\ell}$.

Finally, when $\ell \in \{j, j'\}$, without loss of generality suppose that $\ell = j$. Then the only terms that are not divisible by the monomial axioms as above are $z_{ij}^2 z_{i'j'} + z_{ij'}z_{i'j}^2$. Using the Boolean axioms we can easily convert each such summand to $z_{ij}z_{i'j'} + z_{ij'}z_{i'j}$. The derivation of the sum of these over all $\{i, i'\} \in E(G)$ is analogous, *mutatis mutandis*, to the derivation of (3) above. This completes the proof. □

Proposition 5.5. *The many-one reduction from* Monomial Code Equivalence *to* Tensor Isomorphism *from Grochow & Qiao [GQ21a] is in fact a (2, 4)-many-one reduction.*

Proof. We recall the reduction, then prove that it is a low-degree PC reduction. Let M be a $k \times n$ matrix. We build a 3-tensor of size $(k + 2n) \times n \times (1 + 2n)$ as follows. The first frontal slice is $\begin{bmatrix} M \\ 0_{2n\times n} \end{bmatrix}$. The remaining $2n$ slices all have just a single nonzero entry, which serve to place a 2×2 identity matrix "behind and perpendicular" to M, one 2×2 matrix in each column. Let us index these slices by $[n] \times 2$. Then the (i, b) slice has a 1 in entry $(2(i - 1) + b, i)$, for all $i \in [n], b \in [2]$. Let us call this tensor $r(M)$. Then the reduction maps M, M' to $r(M), r(M')$.

Let X, X', Y, Y', Z, Z' be the variable matrices for the TI equations for $r(M), r(M')$, and let A, B, A', B' be the variable matrices for Monomial Code Equivalence of M, M' (that is, $AMB^T = M'$, A is invertible, B is monomial and invertible). Consider the substitution:

$$X \mapsto A \oplus (B \otimes I_2) \qquad Y \mapsto B \qquad Z \mapsto 1 \oplus (B' \circ B') \otimes I_2$$

$$X' \mapsto A' \oplus (B' \otimes I_2) \qquad Y' \mapsto B' \qquad Z' \mapsto 1 \oplus (B \circ B) \otimes I_2.$$

As before, $B \circ B$ denotes the Hadamard or entry-wise product. Let us see what the TI equations become under this substitution. We get

$$AMB^T = M' \qquad AA' = A'A = \mathrm{Id} \qquad BB' = B'B = \mathrm{Id} \qquad (B'\circ B')(B\circ B) = (B\circ B)(B'\circ B') = \mathrm{Id}$$

Indeed, notice that the effect of the $B \otimes I_2$ in X and the B in Y is that the row and column locations of the 2×2 matrix gadgets get permuted in the same way, and the gadget get multiplied by the *square* of the nonzero entries of B. These are then multiplied by the $B' \circ B'$ in Z.

Now, we derive these equations from the equations for Monomial Code Equivalence. The first three are already present in the equations for Monomial Code Equivalence. The last one we expand out, to see that we need to derive:

$$\sum_j b_{ij}^2 (b'_{jk})^2 = \delta_{ik} (\forall i, k)$$

Now, for $i \neq k$, we may take the equation $\sum_j b_{ij}b'_{jk}$ and square it, to derive

$$\sum_{j\neq j'} b_{ij}b'_{jk} + b_{ij'}b'_{j'k} + \sum_j b_{ij}^2 b_{j'k}^2.$$

Each term in the first sum is divisible by one of the monomial axioms $b_{ij}b_{ij'}$ since $j \neq j'$, and the second sum is what we wanted to derive.

Finally, for $i = k$, we square the equation $\sum_j b_{ij}b'_{ji} - 1$ and add to it $2\left(\sum_j b_{ij}b'_{ji} - 1\right)$. We then proceed to cancel terms with the monomial axioms as above, and end up with $\sum_j b_{ij}^2(b'_{ji})^2 - 1$, as desired. □

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